

LLM-based query expansion

Brief Overview of the Project

Our project's topic will be LLM-based query expansion. During our learning process, we will try to find out whether LLM-based query expansion improves retrieval effectiveness compared to classical retrieval without expansion. Moreover, next to assessing which approach led to selecting more suitable documents, we will conduct a qualitative analysis based on the queries themselves. Testing whether LLM expansion works better for longer or shorter queries, or identifying which LLM generated expansion terms contribute more to retrieval performance are examples of details we will be looking at.

Query Expansion Approaches

In our project, we will evaluate three retrieval configurations:

1. Baseline Retrieval - **BM25** without query expansion
2. Classical Query Expansion - **BM25 + pseudo-relevance feedback**
3. LLM-Based Query Expansion - **Augment** the original query with **LLM-generated terms**

Technical Stack

Our choice of the LLM is **Llama 3.2** because it is the most popular and accessible model to run locally. At first, we want to stick with the aforementioned LLM, seeing whether we could test other models later on in the project.

Our dataset choice is **vaswani**. The reason we wanted to work with it is mostly didactical, since it is a relatively small dataset (around 11000 samples and 93 queries), where indexing is done almost instantly. Besides this, it was part of the introduction to pyterrier provided in this course.

Experimental Procedure

1. Index the dataset using PyTerrier.
2. Run baseline BM25 retrieval.
3. Apply classical query expansion.
4. Generate LLM-based query expansions.
5. Retrieve documents using expanded queries.
6. Evaluate retrieval results using IR metrics.
7. Compare performance across methods.